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A multiplex method for the detection of food microorganisms based on suspension microarrays combined with recombinase-aided amplification

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ABSTRACT

Foodborne diseases caused by foodborne pathogenic microorganisms have long been a severe public health problem. Timely and accurate detection of food-borne microorganisms can minimize their harm and enable the development of efficient control strategies. In this study, we establish a multiplex nucleic acid detection method based on recombinase-aided amplification reaction on a suspension microarray (referred to as SC-RAA) to identify and quantify microorganisms. The experimental results confirmed that this approach has an outstanding sensitivity, with a limit of detection (LOD) of 0.8 fM and a linear range of 0.8 fM–80 pM, spanning five orders of magnitude. Moreover, the prepared method also showed exquisite specificity, which the amplified signal can be detected only when the target nucleic acid is present. The recovery rate of the SC-RAA assay was 82.2%–113.4%, suggesting that the SC-RAA assay is performing well and accurately measuring the concentration of the substance in the sample. In addition, the consistency analysis showed that the results of the SC-RAA detection were also in good agreement with those of traditional qPCR detection ($R^2 > 0.95$). These results highlight the potential of the SC-RAA assay to be applied as a multiplex and sensitive DNA detection platform for identifying and quantifying foodborne pathogenic microorganisms.

1. Introduction

Food microbiological testing is an essential indicator of the quality of food safety, but also a necessary means of judging the food processing environment and food hygiene, which is crucial for food safety (Bao et al., 2022; Rajapaksha et al., 2019). Traditional food microbiological testing methods are based on microbiological culture, which is complicated and takes a long time (microorganisms need to be cultured for at least 48 h) (Chen et al., 2021; Jadhav et al., 2018). In addition, traditional testing methods consume large amounts of media, cultures, and other consumables (Kim et al., 2023). With the development of molecular biology technology, the application of polymerase chain reaction (PCR), and isothermal amplification technology, food microbiological testing is more rapid, accurate, and sensitive, and can detect and deal with food safety problems in a timely manner to protect public health and safety (Hadi et al., 2023; Li et al., 2022; Liu et al., 2019). At the same time, this kind of molecular technology also has the advantages of convenient operation, low cost, and short time, which also makes it has a broader application prospect (Vidyadharani, et al., 2022). However, since food safety problems are usually caused by more than just a single microorganism, ordinary molecular tests cannot detect multiple microorganisms that threaten food safety at one time. Therefore, multiplex assays are needed to achieve rapid, accurate, and efficient detection of foodborne pathogens.

The most common multiplex molecular assay for detecting foodborne pathogens is multiplex PCR (Huang et al., 2021; Villamizar-Rodriguez & Lombo, 2017). However, multiplex PCR has significant limitations. Firstly, this method is limited by the limitation of the instrumentation (limited fluorescence channels of the PCR instrument), which usually can only detect 3–5 species of microorganisms (Aydin et al., 2018; Gu et al., 2019). Secondly, the optimal PCR conditions vary from primer pair to primer pair, e.g., longer fragments require longer extension times, while smaller fragments are always preferentially amplified (Sint et al., 2012). Solid-phase microarray technology is also a

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multiplex method that can detect and identify multiple microorganisms simultaneously, but it is too costly to be widely used (Rasooly & Herold, 2008). With the rapid changes in molecular biology technology, suspension microarray technology has been developed based on fluorescence encoding technology and laser analysis technology (Vafajoo et al., 2018). Different from solid-phase chips, suspension chips are more flexible, more accessible to manipulate, and cheaper (Lin et al., 2016). Although suspension microarrays allow direct multiplex detection by hybridization of capture probe attached to fluorescence-encoded microspheres to the DNA of target foodborne pathogens, sensitivity is limited (Janse et al., 2012; Miller & Tang, 2009). Many low-abundance foodborne pathogens are difficult to detect by weak nucleic acid hybridization signals. Therefore, before performing nucleic acid hybridization experiments using suspension microarrays, target fragments are usually amplified by PCR or isothermal amplification, which undoubtedly makes the detection process more cumbersome (Sun et al., 2014). In this study, we would like to develop a more straightforward method to achieve high throughput and sensitivity for detecting foodborne pathogens.

Recombinase-aided amplification (RAA) is a nucleic acid detection technique that is considered an alternative to PCR (Shahrajabian et al., 2021). RAA is an isothermal nucleic acid amplification technique that amplifies target gene fragments using sample DNA as templates for 15–40 min under isothermal conditions (typically 37–42 °C) (Hu et al., 2020; Leonardo et al., 2021; Lobato & O'Sullivan, 2018). This method has been applied in many fields due to its simplicity, low instrument dependency, sensitivity, and rapidity (Wu et al., 2021). In this study, we developed a multiplex nucleic acid detection method based on RAA reaction on a suspension chip (referred to as SC-RAA) for the multiplex quantitative detection of microorganisms responsible for foodborne illnesses, including *Listeria monocytogenes*, *Vibrio parahaemolyticus, Aspergillus niger*, and *Salmonella typhimurium* (Gil-Serna et al., 2019; Zhang et al., 2015). This multiplex approach can be used to rapidly detect and analyze foodborne pathogens and provide an essential basis for the prevention and control of foodborne microbial diseases.

2. Materials and methods

2.1. Acquisition of DNA fragments of foodborne pathogens

We synthesized conserved gene fragments from four foodborne microorganisms (*L. monocytogenes*, *V. parahaemolyticus*, *A. niger*, and *S. typhimurium*) and inserted them into the plasmid, pUC57, making recombinant plasmids. Then, these contained target genes recombinant plasmids were transferred to *E. coli* DH5α for amplified culture, and plasmid DNA containing the target gene fragments was obtained by Plasmid Midi Preparation Kit purchased from Beyotime Biotechnology Co., Ltd (Shanghai, China), which was used as DNA templates for the SC-RAA assay. A Nanodrop 2000 (Thermo Fisher Scientific, Wilmington, DE, USA) was used to measure the concentrations and purity of DNA.

2.2. Primers design of the SC-RAA assay

The genomic information of foodborne pathogens was obtained from the National Center for Biotechnology Information (NCBI) database: M90846.1 (*S. typhimurium*), MH593586.1 (*A. niger*), *L. monocytogenes* (M24199.1), and *V. parahaemolyticus* (MF983554.1). Different from the design of regular PCR, the length of RAA primers is generally 30 to 35 nucleotides, and the primers that are too short will seriously affect the activity of recombinase. According to the obtained specific gene sequences and RAA primer design principles (Daher et al., 2016; Xiang, Diao, et al., 2022), RAA primers were designed by Primer Premier 5.0 software (Premier Biosoft, Palo Alto, CA, USA) and verified for non-specificity and sensitivity. All the primers, including biotin-labeled primer and fluorophore-labeled primers, were listed in Table 1, and synthesized by Sangon Biotech (Shanghai) Co., Ltd (Shanghai, China).

2.3. Examination of the thermal stability of avidin-modified fluorescence-encoded magnetic microbeads (AFMMs)

AFMMs were provided by Beadstar Biotechnology (Hainan Province) Co., Ltd. (Haikou, Hainan province, China) The fluorescent-coded microspheres we used were prepared by incorporating two fluorescent substances into the microspheres. Due to the varying proportions of fluorescent dyes within each set of microspheres, they display distinct fluorescence intensities in the APC and APC-Cy7 channels when excited by a 640-nm wavelength. The excitation and emission wavelength information of the APC and APC-Cy7 channels are listed in Table S1. This unique fluorescence property enables clear differentiation of the four microsphere groups in the fluorescence plot of the flow cytometer's APC/APC-Cy7 channels, forming four separate clusters of microspheres. To determine whether the AFMMs can tolerate the temperature of RAA reaction, the AFMMs were incubated at constant temperature for 1.5 h at 39, 45, 51, and 57 °C, respectively. Test probe, 5′ biotinylated as well as 3′ FAM-labeled nucleic acid sequences, was used to test the efficiency of primer attachment to the AFMMs.

2.4. Optimization of conditions for primer attachment to AFMMs

The concentration of the primer attachment to the AFMMs was optimized as follows: 1.8×10^5 of AFMMs were dispersed in 90 μL of 1 × TBST solution, and 10 μL each of the test probe with primer concentrations of 62.5, 125, 250, 500, and 1000 nM, respectively, were added. The mixture was stirred for 120 min at a temperature of 25 °C using a temperature-controlled vortex mixer at a speed of 2000 rpm. Then, the AFMMs conjugated with the fluorescent probe were washed three times with 100 μL of 1 × TBST solution and finally resuspended with 400 μL of 1 × TBST solution. Fluorescent signals of FAM-labeled test probe attached to the AFMMs were measured by flow cytometry. Next, we optimized the time of primer attachment to the AFMMs. The test probe was incubated with AFMMs for 30, 60, 90, and 120 min, respectively. Then, the AFMMs conjugated with the fluorescent probe were washed

Table 1
Oligonucleotide sequences of primers and designed for the SC-RAA assay.

Species	Sequence(5′–3′)	Product
L. monocytogenes	F: Biotin-TATATATACATTAACACTCAGCATTGATTTGCCAGGTA R: GATTCACTGTAAGCCATTTCGTCATCATAA	193 bp
V. parahaemolyticus	F: Biotin-TATATATATGATCCAAATAGTAATTCGCTCGCTGACCAAC R: ATCCTTCAACATCTTACGCAGAGTAGAAATCG	220 bp
A. niger	F: Biotin-TATATATACCTACCTCCGCCATTTCGCCTACGACCCTC R: CTCCTTCGGCATCATGGAGCAAGTACCCAC	254 bp
S. typhimurium	F: Biotin-TATATATATCTCTATTGTCACCGTGGTCCAGTTTATCGT R: AACTTCATCGCACCGTCAAAGGAACCGTA	223 bp
Test probe	Biotin-TGATCCAAATAGTAATTCGCTCGCTGACCAAC-6-FAM	N/A

F: forward primer; R: reverse primer; N/A: not applicable. The test probe is a 5′ biotin-modified and 3′ fluorescently labeled DNA which is used to explore the conditions for linking microspheres with primers through the avidin-biotin system.

three times with 100 μL of 1 × TBST solution and finally resuspended with 400 μL of 1 × TBST solution. Similarly, the fluorescence signal of the test probe attached to the AFMM was finally detected by flow cytometry.

2.5. Establishment of the SC-RAA assay

RAA Nucleic Acid Amplification Kit was purchased from Jiangsu Qitian Gene Biotechnology Co., Ltd (Wuxi, Jiangsu province, China). The 50 μL system contained 6 μL of suspensions (1×10^4 upstream primer attached-AFMMs), 2.5 μL of downstream primer (10 μM), 5 μL of 1 mM iFluor 488 labeled-dUTP (prepared according to established previously in our lab (Liu et al., 2023), 4 μL of plasmid DNA template, and other components of enzymatic reactions were performed at a single temperature (39 °C) for half an hour. The APC and APC-Cy7 channels of the flow cytometer were used to determine the fluorescence coding of the four AFMMs. The FITC channel then detected the fluorescent signals of the amplified products.

2.6. Optimization of the SC-RAA assay

Different lengths of DNA-arms (consisting of 0, 4, 8, and 12 bases, respectively) were inserted at the beginning of the primer. 1 nM of *V. parahaemolyticus* as a template was added to the RAA reaction system containing primer-attached AFMMs. After the isothermal amplification process, the AFMMs with amplification products attached were washed three times with 1 × TBST solution and then resuspended with 100 μL of 1 × TBST solution. The SC-RAA assay was performed at 35, 37, 39, 41, and 43 °C, respectively, to clarify the optimal reaction temperature. We set up six gradients (10, 20, 30, 40, 50, and 60 min) at 10 min intervals to find the most appropriate reaction time. The SC-RAA assay was performed using 1 nM of *V. parahaemolyticus* as the template. All the tests were conducted three times. Finally, the concentration of Mg^{2+} in the reaction system was optimized. We performed the SC-RAA assay in the reaction system containing Mg^{2+} concentrations of 7, 14, 21, and 28 mM, respectively. After the reaction, the amplification products bound to AFMMs were washed with 100 μL of 1 × TBST solution, and the fluorescence intensity of the amplified products was detected by flow cytometry. After the reaction, the AFMMs with amplification products were washed with 100 μL of 1 × TBST solution, and the fluorescence intensity of the amplified products was detected by flow cytometry.

2.7. Performance evaluation of the SC-RAA assay

To assess the sensitivity of the SC-RAA assay, we sequentially diluted the target DNA in a 10-fold concentration gradient. No template served as a negative control. The optimized SC-RAA detection conditions were employed for amplification, and all tests were conducted in triplicate. Four distinct encoded AFMMs congregated with specific upstream primers, and corresponding downstream primers are added to the SC-RAA reaction system. The SC-RAA assay's specificity underwent assessment through comparing the fluorescence signal intensity of the amplified product from the corresponding AFMMs of the target with that from other non-corresponding AFMMs. All tests were performed in triplicates. We then evaluated the detection performance of the SC-RAA assay in simulated actual samples. The bacterial solution was added to milk and diluted sequentially into 1×10^6, 1×10^5, 1×10^4, 1×10^3, 1×10^2, 1×10^1 and 1 CFU/mL. Next, 5 mL of contaminated milk was taken from each gradient to extract the DNA, respectively. After the SC-RAA reaction, we measured the fluorescence intensity of the amplified product on the AFMMs and calculated the standard curves.

2.8. Comparison of results between the real-time fluorescence quantitative PCR (qPCR) and the SC-RAA assay

To determine whether the results of the SC-RAA test were consistent with those of other methods, we conducted a correlation analysis between the outcomes of the SC-RAA assay and qPCR testing. qPCR reactions are performed using the fluorescent quantitative detection reagent kit (Tiangen Biotech (Beijing) Co., Ltd. Beijing, China). Each 20 μL amplification system contains the following components: 0.6 μL of 10 μM forward primer, 0.6 μL of 10 μM reverse primer, 10 μL of 2 × fluorescent quantitative PCR premix, 1 μL of plasmid DNA template, and 7.8 μL of ddH_2O. The reaction program is carried out in two steps according to the instruction manual: pre-denaturation at 95 °C for 180 s, 1 cycle; denaturation at 95 °C for 5 s, annealing/extension at 60 °C for 15 s, 40 cycles in total. We calculated the correlation coefficient between the two methods using Origin 2021 software.

2.9. Recovery experiments

We evaluated the accuracy and reliability of the SC-RAA assay by recovery experiments. We used milk contaminated with three concentrations of bacteria as test samples (high concentration (1×10^6 CFU/mL), medium concentration (1×10^4 CFU/mL), and low concentration (1×10^2 CFU/mL)). After DNA extraction, the amplification was performed using the SC-RAA method. Recovery was calculated according to the formula (the measured bacterial concentration/the incorporated bacterial concentration) × 100%.

2.10. Statistical analysis

The flow cytometer data files of the SC-RAA experiment were analyzed using BD Accuri™ C6 Plus software. Forward scatter (FSC) and side scatter (SSC) channels were used to gate the total microsphere population. APC and APC-Cy7 fluorescence channels were employed to distinguish magnetic fluorescence-encoded microsphere populations. The FITC channel measured the fluorescence intensity of the products on the magnetic fluorescence-encoded microspheres, which was used to determine the sample concentration. The linear regression was performed using Origin 2021 software to establish a linear model correlating concentration with fluorescence intensity. The specificity graph was generated using GraphPad Prism 10.

3. Results and discussion

3.1. Mechanism of the SC-RAA assay based on flow fluorescence techniques

Microbiological contamination of food has emerged as a significant problem in food safety and public health, and the presence of microorganisms in food may lead to foodborne diseases, posing a potential threat to human health (Pissuwan et al., 2020). In this study, we develop the SC-RAA assay for multiplex detecting food microorganisms based on suspension microarrays combined with recombinase-aided amplification. The mechanism of the SC-RAA assay is shown in Fig. 1. Prior to the start of the SC-RAA reaction, primers of the four foodborne pathogens were attached to four different fluorescence-encoded AFMMs *via* the interaction of avidin and biotin. In the RAA reaction system, the recombinant enzyme, single-stranded binding protein, and primers (including the upstream primer attached to the AFMMs and the free downstream primer) form a complex scanning the target DNA and unwinding the double-stranded DNA at sequences homologous to the primer. The single-stranded binding protein (SSB) prevents the single-stranded DNA from replicating, and the strand extension is accomplished by the DNA polymerase in the presence of energy and dNTP. The fluorescence of the amplified product is achieved by replacing dTTP with partially iFluor 488-labeled dUTP during the reaction. Finally, the species of food microorganisms was clarified by detecting the fluorescence coding on the AFMMs (APC and APC-Cy7 channels), and the concentration of the substance was measured by detecting the fluorescence intensity of the amplification product (FITC

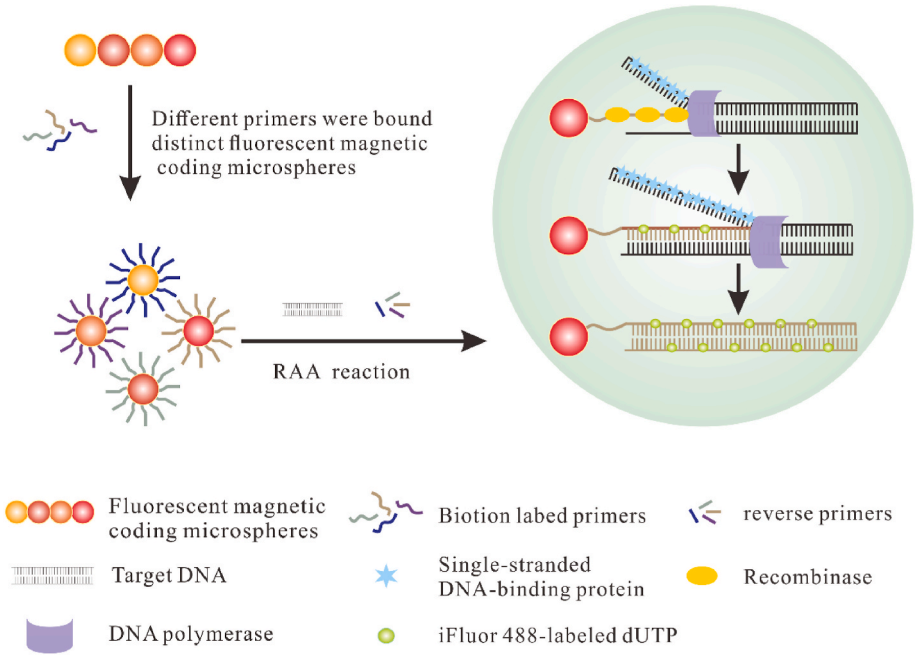


Fig. 1. Schematic illustration of multiple detection of foodborne pathogens based on the SC-RAA assay.

channel). Unlike multiplex PCR assays, which have a minimal number of items (limited to fluorescent channels) (Duncan et al., 2019; Hodzic et al., 2023), the SC-RAA assay can detect up to 100 items, depending on the code of the AFMMs, making it ideal for the simultaneous detection of a wide range of food microorganisms.

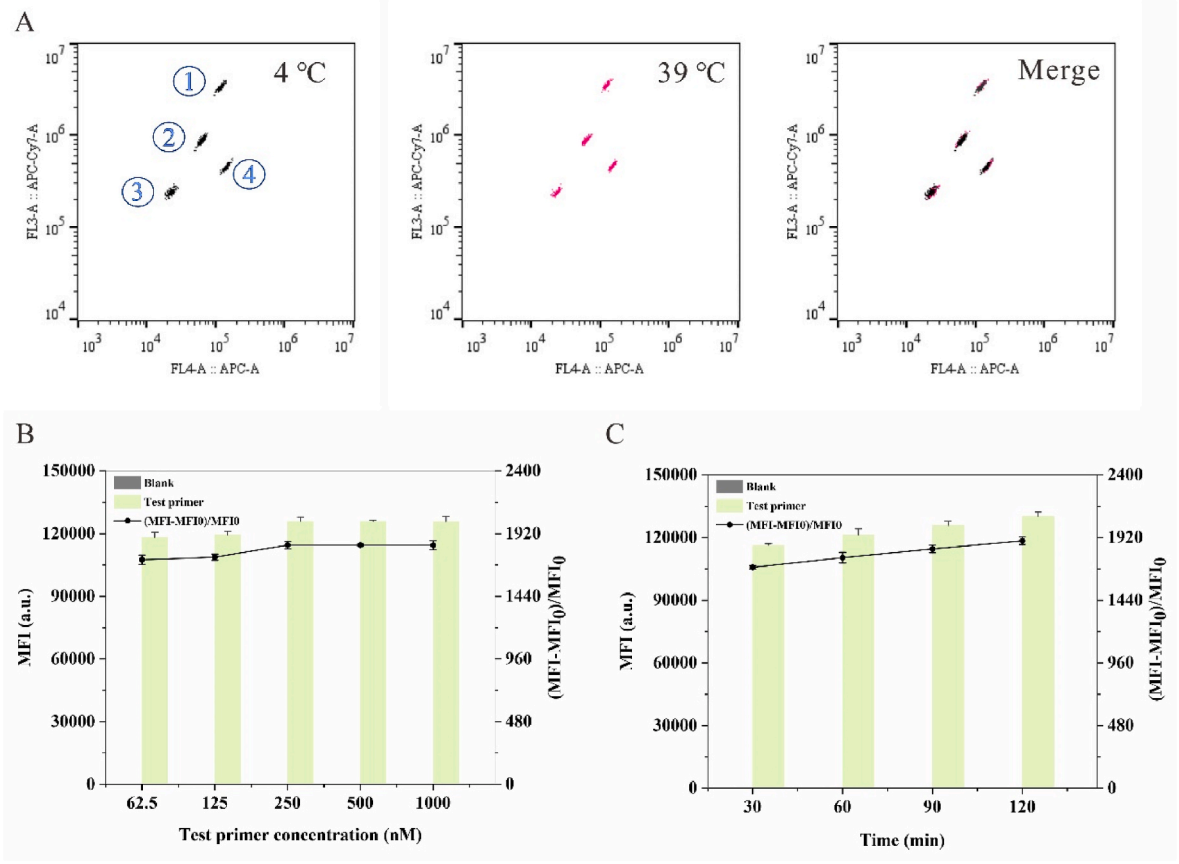


Fig. 2. (A) The effect of temperature on the encoding position of AFMMs. The corresponding detection targets for AFMM ① to ④ are *L. monocytogenes*, *V. parahaemolyticus*, *A. niger*, and *S. typhimurium*, respectively. (B) AFMMs coupled with test probe at different concentrations, and (C) AFMMs coupled with test probe at different times. Here, MFI refers to the measured fluorescence intensity at a specific time point, MFI0 represents the initial fluorescence intensity.

3.2. Establishment of SC-RAA assay

Different from PCR reactions involving multiple rounds of thermal cycling, RAA reactions require lower and more stable temperatures (37–42 °C). However, it still necessitates a certain level of thermal stability in the AFMMs. In this experiment, the selected four AFMMs were incubated at constant temperature for 1.5 h at 39, 45, 51, and 57 °C. As can be seen from the flow diagrams (Fig. 2A and Fig. S1), even at the highest temperatures, these AFMMs' positions hardly change compared to their original positions. The results of thermal stability tests confirm that these AFMMS are fully capable of being used as suspension carriers for isothermal amplification. Subsequently, the AFMMs were immersed in solutions of varying concentrations of the test probe (the test probe was modified with biotin at the 5′ end and labeled with FAM at the 3′ end) to determine the most effective concentration of primers for attachment. The highest value of (MFI-MFI0)/MFI0 was observed when the primer concentration reached 250 nM (Fig. 2B, MFI refers to the measured fluorescence intensity at a specific time point, MFI0 represents the initial fluorescence intensity). Therefore, this concentration was utilized for all subsequent experiments. Additionally, the impact of ligation time on the attachment of primers to AFMMs was investigated. The fluorescence of the test probe attached to AFMMs increased with ligation time. To minimize the experimental manipulation time, a ligation time of 120 min was selected for subsequent experiments, even though a 2-h ligation duration may not fully optimize the ligation of primers to AFMMs (Fig. 2C).

3.3. Optimization of SC-RAA reaction conditions

The design of reaction primers is the most important intrinsic factor in optimizing the efficiency of the SC-RAA reaction (Li et al., 2019). Here, we designed multiple pairs of primers for target DNA amplification and finally chose four pairs of primers with high amplification efficiency and non-specific amplification as the SC-RAA primers. Unlike the reaction in solution, the SC-RAA is a liquid-phase microsphere suspension microarray, and we need to consider the spatial site-blocking effect caused by the AFMMs (Liu et al., 2023). In this work, we designed spacer-arms of different lengths at the 5′ end of the upstream primer. When the arm length is eight bases, the amplified fluorescence signal intensity is the highest, and the value of (MFI-MFI0)/MFI0 is also the highest (Fig. 3A). Therefore, we chose the 8-base DNA arm at the 5′ of the upstream primer in the following experiments. Next, we investigated the effect of reaction temperature on the amplification efficiency of SC-RAA. When the temperature was less than 39 °C, the fluorescence intensity of the amplification products rose as the temperature climbed. However, when the temperature rose over 39 °C, the fluorescence intensity of the amplified products dropped. The results of the temperature range screening revealed that the primer pairs had a satisfactory detection effect at 39 °C (Fig. 3B). In liquid phase systems, the RAA often achieves sufficient amplification product within a short time frame of 5–20 min (Xiang, Shang, et al., 2022; Xue et al., 2020; Zhang et al., 2017). However, the RAA reaction on magnetic fluorescence-encoded microspheres to obtain a sufficient amount of amplification product is something that we need to test further. In this study, the SC-RAA reaction was carried out continuously for 10, 20, 30, 40, 50, and 60 min to

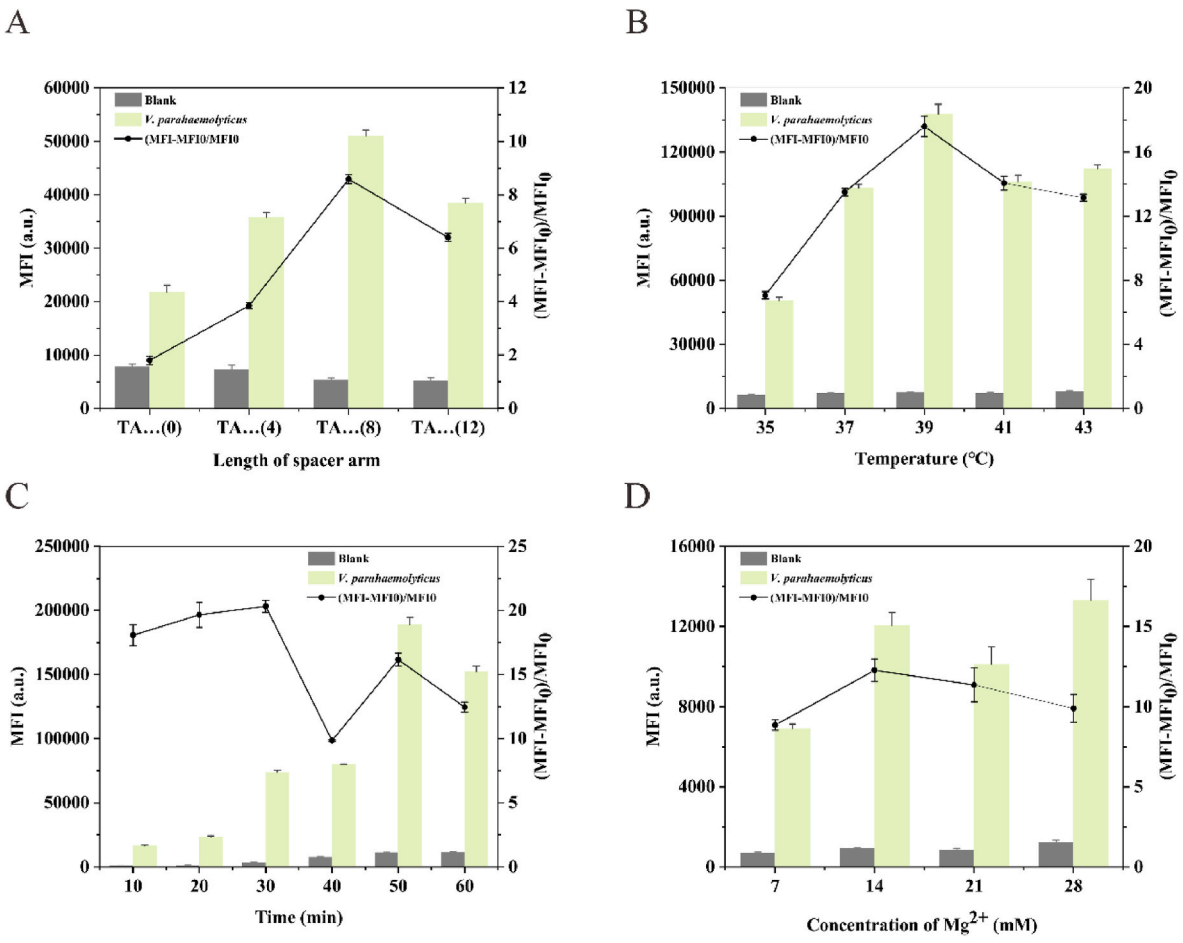


Fig. 3. Optimization of reaction conditions of SC-RAA assay. The screening of (A) the length of the spacer arm, (B) amplification temperature, (C) amplification time, and (D) concentration of Mg^{2+}.

determine the optimal time required for the SC-RAA reaction. The fluorescence intensity of the amplification products increased with the duration of the reaction. However, it should not be overlooked that the fluorescence of the blank control group increased as well. After analyzing the experimental results, we concluded that a reaction time of 30 min provides the optimal signal-to-noise ratio (Fig. 3C). Subsequently, we investigated the impact of varying Mg^{2+} concentrations on the SC-RAA reaction. When the concentration of Mg^{2+} was 14 mM, it had the maximum signal-to-noise ratio, so we chose 14 mM concentration of Mg^{2+} as the working concentration (Fig. 3D).

3.4. Sensitivity of the SC-RAA assay

After optimizing the assay conditions, we further examined the assay performance of the SC-RAA assay. To investigate the detection range of the SC-RAA assay, amounts of templates were prepared by continuous dilution. The linear response (standard curve) of the four foodborne microbial DNA measuring by SC-RAA assay. The findings demonstrated that the four foodborne microbial DNAs had a broad linear detection range (0.8 fM to 80 pM) with high correlation coefficients ($R^2 > 0.98$) for the SC-RAA assay (Fig. 4A–D). The limit of detection (LOD) of the four foodborne microbial DNA, in terms of 3 times deviation over the blank response rule, is calculated as 0.8 fM (Wang, et al., 2020). This significantly lower value of the LOD and broader linear range, spanning five orders of magnitude, confirm the superior analytical performance of the SC-RAA assay.

3.5. Specificity of the SC-RAA assay

In this study, the selectivity of the SC-RAA assay was evaluated by measuring its anti-interference ability for irrelevant targets. Four plasmids containing DNA fragments from various foodborne microorganisms were sequentially introduced to four test tubes each equipped with four distinct coding AFMMs attached to specific upstream primers and free downstream primers, along with RAA reaction reagents. The results were illustrated in Fig. 5A–D, wherein the corresponding AFMMs exhibited very strong fluorescence signals in the presence of specific target DNA. In contrast, the other non-corresponding AFMMs showed negligible signals. These findings demonstrated that the SC-RAA assay has high selectivity for the target DNA.

3.6. Analytical application for the determination of four foodborne microbial DNA in milk

To evaluate the performance of the SC-RAA in the detection of natural food contamination samples, we utilized milk that was infected with *E. coli* containing the target bacterial fragment plasmid as a simulated sample in the analysis (excluding *V. parahaemolyticus*, as it is not commonly found in dairy products). The fluorescence signal of the artificially contaminated sample increased with increasing bacterial concentration. Moreover, the results exhibited that the SC-RAA method achieved a wide detection range in detecting contaminated milk with the target bacteria, with a good linearity in the bacterial concentration range of $1 \times 10^1 - 1 \times 10^6$ CFU/mL (Fig. 6). These findings indicated that the detection method proposed in this study has good analytical performance when simulating the detection of actual samples and can theoretically be used to detect factual samples.

3.7. Comparison with the qPCR assay and other method

To assess the comparability of data obtained from the SC-RAA assay with those from the qPCR assay, we conducted parallel detection using both the SC-RAA assay and the PCR assay on identical samples. Our findings revealed that the detection linear ranges of these two methods are virtually identical. Additionally, linear regression analysis of the assay data from both methods demonstrated strong consistency, with an $R^2 > 0.95$ (Fig. 7). Compared to the newly developed foodborne pathogen detection technologies in recent years (Table 2), the SC-RAA

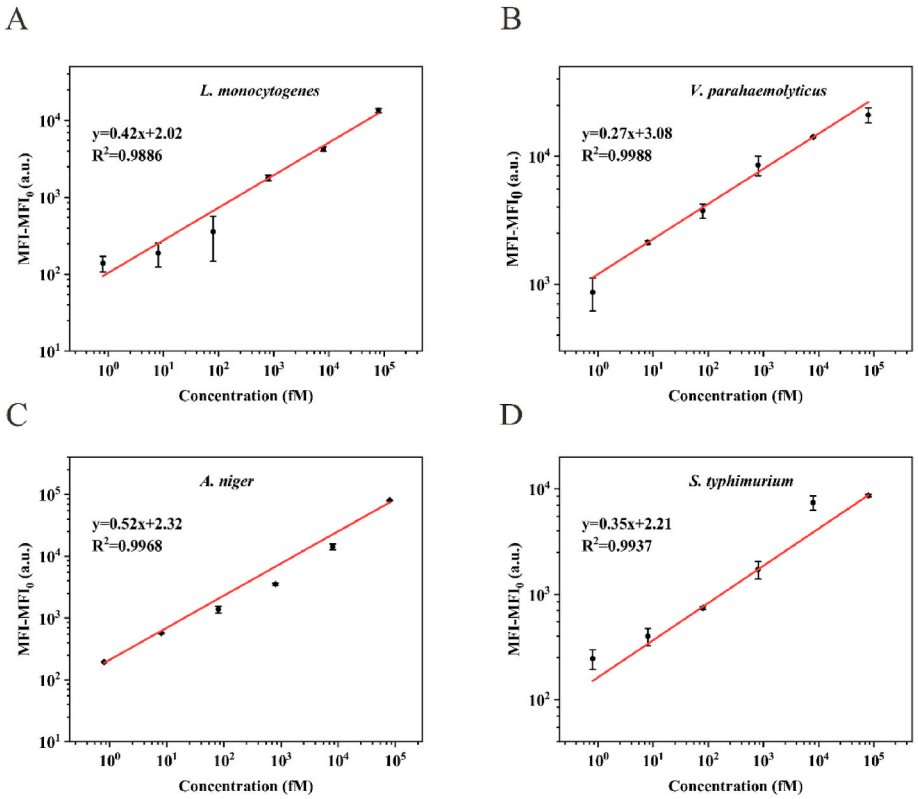


Fig. 4. The linear range of (A) *L. monocytogenes*, (B) *V. parahaemolyticus*, (C) *A. niger*, and (D) *S. typhimurium* by SC-RAA assay.

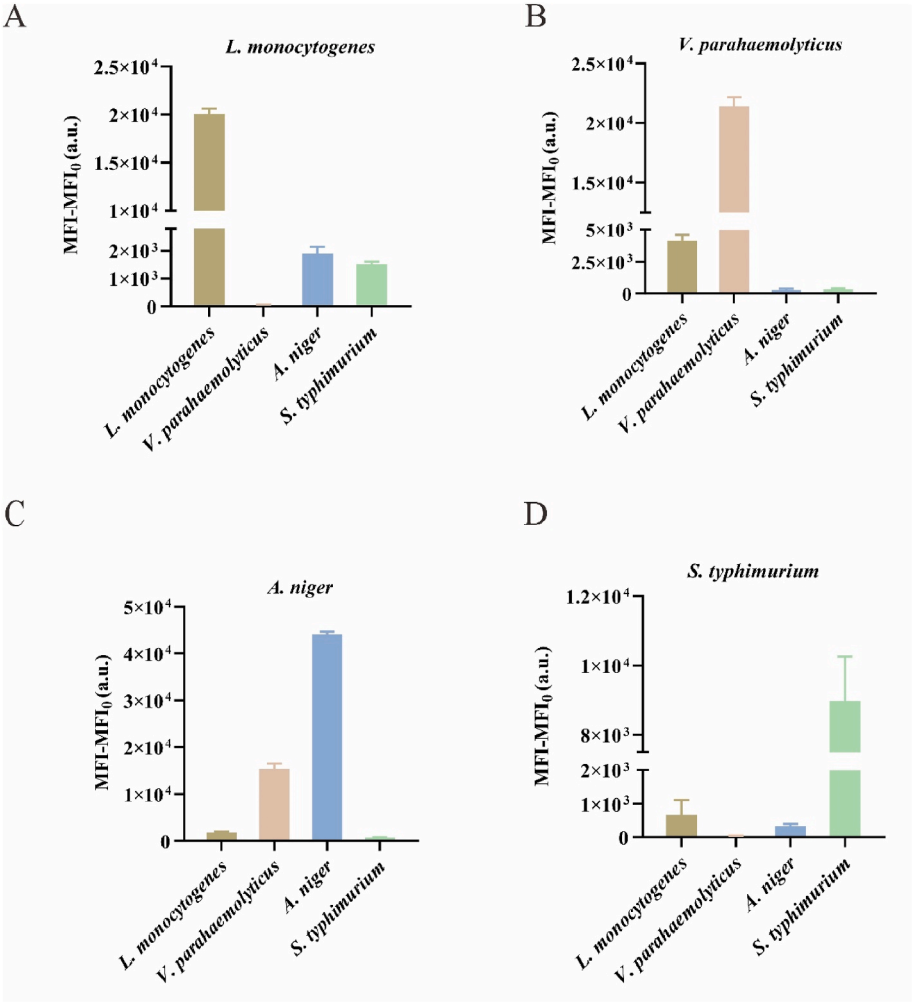


Fig. 5. The specificity of (A) *L. monocytogenes*, (B) *V. parahaemolyticus*, (C) *A. niger*, and (D) *S. typhimurium* by SC-RAA assay.

demonstrates strong comprehensive capabilities. It not only enables simultaneous detection of multiple pathogens but also possesses a wide detection range.

3.8. Recovery experiment

In scientific research, a recovery experiment refers to a type of experiment that aims to determine the ability of a detection method or procedure to recover a known quantity of a substance or material from a complex matrix. A recovery rate of 80%–120% is generally considered acceptable. The data in Table 3 showed that the recovery rate of the SC-RAA assay was 82.2%–113.4%, suggesting that the SC-RAA assay performs well and accurately measures the concentration of the substance in the sample.

4. Conclusion

Food safety has always been an issue of public concern. Microorganisms have the potential to contaminate food throughout production, transportation, and sales. Once contaminated, they will multiply in large quantities, leading to food spoilage and causing foodborne contamination and food poisoning. To prevent foodborne illnesses, it is crucial to employ appropriate and efficient techniques for identifying microorganisms in food. In this study, we developed the SC-RAA assay for multiplex quantitative detection of food microorganisms. The results showed that this method has excellent detection performance, having a wide linear detection range as well as reasonable specificity. It is worth noting that, when compared to traditional qPCR tests, the SC-RAA assay not only detects multiple templates simultaneously, but it is also very fast (approximately 30 min) and operates at a constant temperature of 39 °C, effectively reducing procedure time and cost as well as instrument dependency. In addition, the results of the SC-RAA assay were also in good agreement with those of traditional qPCR detection. In summary, we have developed a multiplex detection method for rapid detection and analysis of foodborne pathogens, providing great convenience for the prevention and control of foodborne microbial diseases. With the increasing prevalence of flow cytometers, this method holds significant application value. To better promote this approach, we may consider integrating a constant-temperature amplification module into the flow cytometer equipment, enabling the integration of nucleic acid amplification and detection. This would effectively streamline the operational process and enhance detection efficiency.

CRediT authorship contribution statement

Wentao Zeng: Investigation, Formal analysis, Data curation. **Xihao Fan:** Investigation, Formal analysis. **Azadeh Nilghaz:** Writing – review & editing. **Gengfen Zhao:** Formal analysis. **Lingfeng Yan:** Investigation. **Zhuoxuan Lu:** Writing – original draft, Funding acquisition, Conceptualization. **Liming Zhang:** Supervision, Project administration, Funding acquisition, Conceptualization.

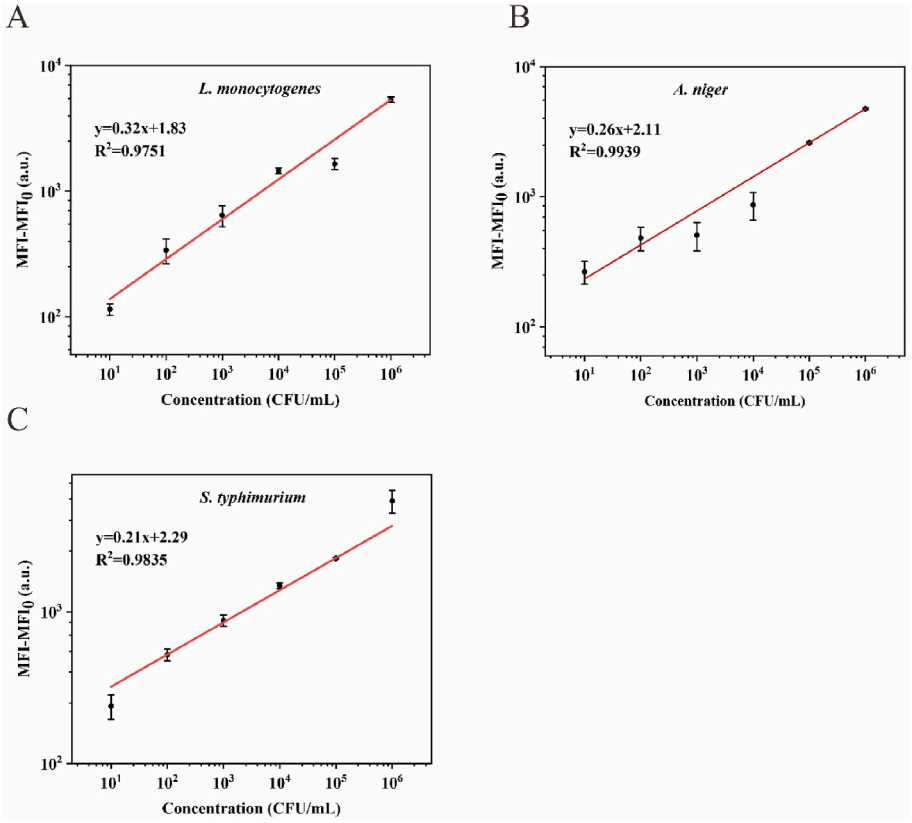


Fig. 6. The sensitivity detection of (A) *L. monocytogenes*, (B) *A. niger*, and (C) *S. typhimurium* in milk based on SC-RAA assay.

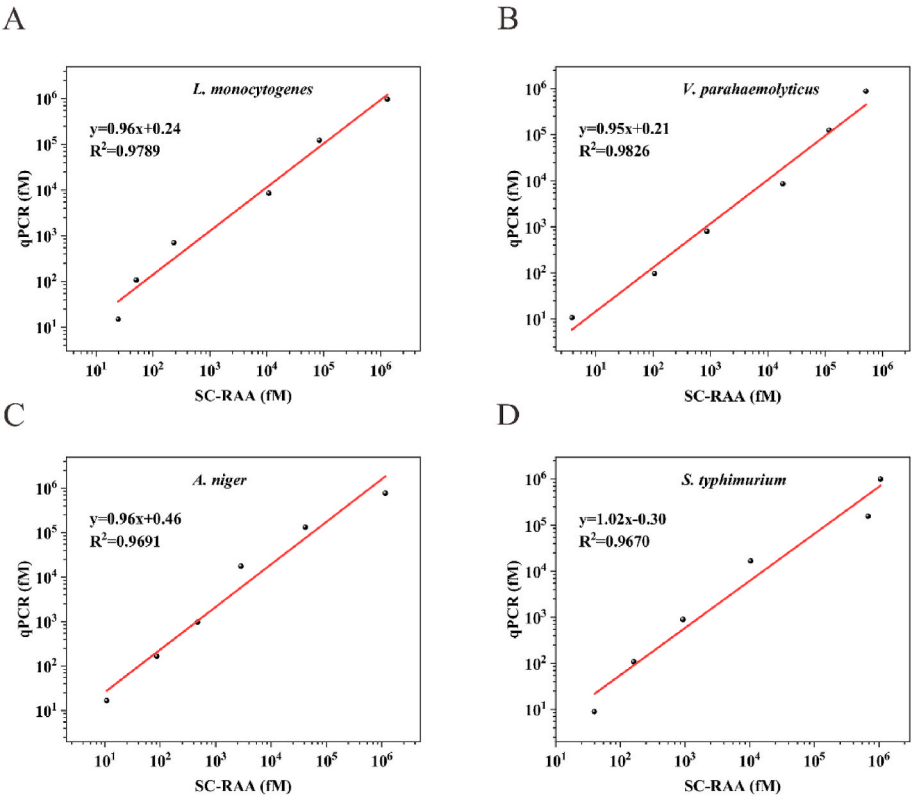


Fig. 7. Comparison of the proposed approach and qPCR method for detecting (A) *L. monocytogenes*, (B) *V. parahaemolyticus*, (C) *A. niger*, and (D) *S. typhimurium* in different DNA concentrations.

Table 2
Comparison of the SC-RAA assay with prior methods.

Method	Assaying time (h)	Linear range CFU/mL	capability for multiplex detection	Reference
SC-RAA	0.5	$10–10^6$	Yes	This work
CMORA	2	$10–10^5$	No	Ren et al. (2022)
Mag-Cds-MPCR-CE	2.5	$10–10^5$	Yes	He, et al. (2022)
Colorimetric aptasensor	2.4	$5.4 \times 10^3–10^8$	No	Zhang et al. (2016)
SERS-based biosensor	0.5	$19–1.9 \times 10^6$	Yes	Liu et al. (2017)

Table 3
The recoveries of *L. monocytogenes*, *A. niger*, and *S. typhimurium* from milk at different concentration levels (n = 3).

Species	Added (CFU/mL)	Found (CFU/mL)	Recovery (%)	RSD (%)
L. monocytogenes	100	88	88.00	2.41
	10000	10231	102.31	8.06
	1000000	903900	90.39	1.81
A. niger	100	113	113.40	7.72
	10000	8651	86.51	7.37
	1000000	890100	89.01	3.01
S. typhimurium	100	82	82.20	6.92
	10000	8887	88.87	5.15
	1000000	877400	87.74	9.25

RSD: relative standard deviation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fbio.2024.104365.

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